Data cleaning

Lesson objectives

By the end of this session, you'll be able to:

- Describe the importance of data cleaning for the research data lifecycle
- Identify data cleaning needs for an example dataset
- Use the pipe operator and functions from the tidyverse to clean data of an example dataset
- Write reproducible code by applying concepts of literate programming

Activity

Open the survey dataset, and discuss the following questions:

- Is this dataset in a consistent format for subsequent work and analysis? Why or why not?
- What fields would you want to use to answer the project's research question? Do they exist already?
- Are there any fields that you think might be tricky to work with? Why?
- What fields would you want to modify or correct in this dataset? How would you do that?

Make a list of things you would want to modify, transform or correct in this dataset, and choose someone in your group to share the list with the larger group.

You will have 20 min in the break out room.

Breakout!

Data cleaning

- Data cleaning is the process of checking and fixing your data to ensure it is ready for analysis
- It is time consuming but crucial ("garbage in, garbage out")
- Documenting your data cleaning process ensures reproducibility

Good practices for data cleaning

- Do not directly edit raw data
- Data cleaning scripts:
  - Inputs raw data
  - Performs data cleaning
  - Outputs the cleaned dataset
- Always share clean versions of the data

Data cleaning steps

Not all steps will apply to all datasets, always refer back to your data and research questions

- Fix column names
- Remove duplicate observations
- Correct spelling mistakes
- Standardize letter case and use of space or other special characters
- Convert variables to correct variable types (e.g., numeric, categorical, date, etc).
- Group values into larger categories
- Merge and split columns
- Create new variables
- Deal with missing values
- Subset data
- Reshape dataset for analysis

The pipe operator

- Symbol |> (or %>%)
- Feeds the result of one function directly into the other function
- Functions well with the tidyverse syntax for data cleaning
- Example:

```
print(count(survey_data, Age, Gender), n = Inf)


count(survey_data, Age, Gender)|>
    print(n = Inf)
```

Time to stretch!
(10 min break)

Time to clean!

Optional activity

- Practice your skills in data cleaning with another dataset!
  - Squirrels in Central Park, in New York City
- Not required for the course
- Instructions on the bottom of Day 4 - Lesson 1 content
  - Download the dataset
  - Perform requested data cleaning actions
  - Save the cleaned dataset
  - Can start from scratch or use the fill-in-the-blanks template